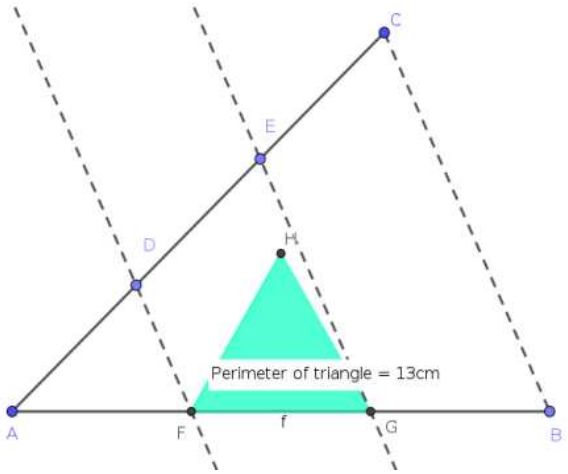
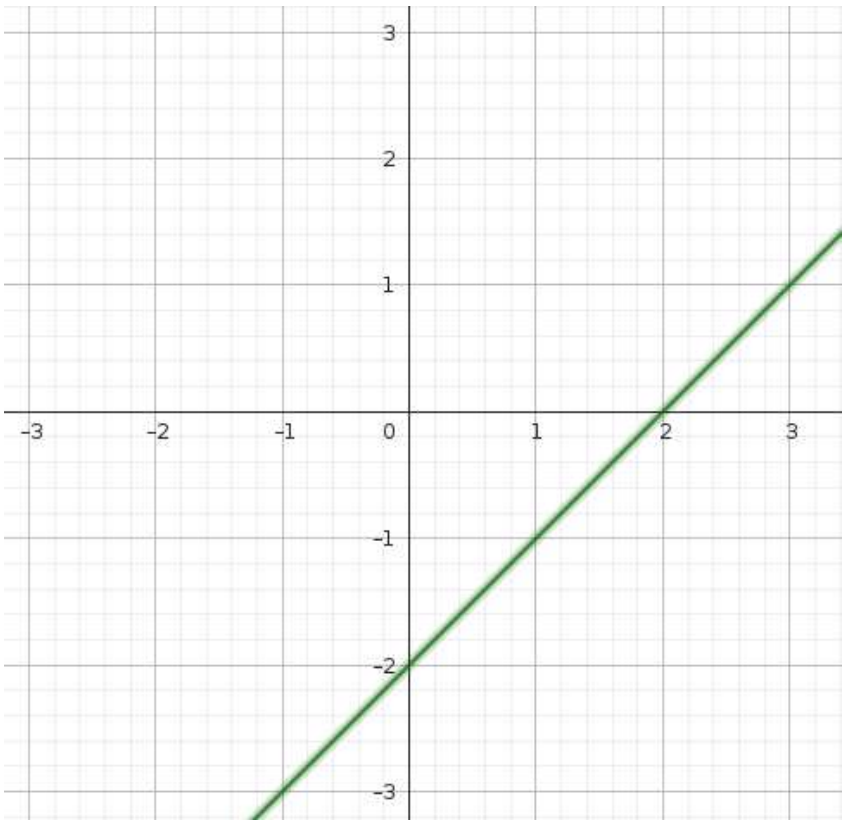
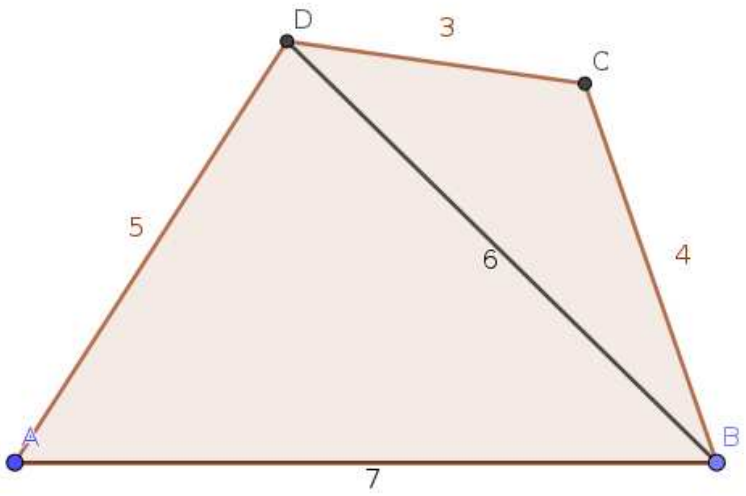
Qn No	Sub Qn	Answer	Score
1		$Mean = \frac{70+65+45+54+72+66+58+50}{8} = \frac{480}{8} = 60$	2
2		x+y = 21 ; x- y = 5 x = 13 , y = 8	2
3		$\pi r^2 = 4\pi$ r = 2	2
4		$Unshaded\,region = \frac{360-120}{360} = \frac{2}{3}$	2
5	a	BC = $\sqrt{3}$cm	1
	b	Perimeter = $4\times\sqrt{3} = 4\sqrt{3}$cm	2
6	a	Base area= $\frac{6\sqrt{3}}{4}\times 8^2 = 96\sqrt{3}\,cm^2$	1
	b	Volume = $96\sqrt{3}\times 15 = 1440\sqrt{3}\,cm^3$	2
7	a	$\frac{AB\sqrt{3}}{2} = 2\sqrt{3}$ AB = 4 cm	1
	b	Area of triangle ABC $= \frac{\sqrt{3}\times 4^2}{4} = 4\sqrt{3}\,cm^2$	2
8	a	Length = $\frac{12}{4}$= 3 cm	1
	b	Length of the arc = $\frac{3}{2}\times 3 = 4.5\,cm$	2

9	a		3
10	a	$2\pi r$	1
	b	Yes ; 2π	2
11	a	$\frac{A_1}{A_2}=\frac{2\pi r_1\times h_1}{2\pi r_2\times h_2}=\frac{r_1\times h_1}{r_2\times h_2}=\frac{1\times 4}{2\times 3}=\frac{2}{3}$	2
	b	$\frac{30}{S_2}=\frac{2}{3}$ $S_2 = 45\ cm^2$	2
12	a	$P(0) = 2$	1
	b	$P(1) = 6$	1
	c	$\frac{P(1)+P(-1)}{P(0)}=\frac{6+0}{2}=3$	2
13	a	Radius of the large semicircle = 4 cm Radius of the small semicircle = 2cm	1
	b	Area of the large semicircle = $\frac{\pi}{2}\times 4^2=8\pi cm^2$	1
	c	Area of the shaded region $= 8\pi-2(\frac{1}{2}\pi 2^2)=4\pi$	2
14	a	$\|x+2\|=5$ x = 3 or x = -7	2
	b	$\|x-6\|=\|x-2\|$	2

		$x=\frac{6+-2}{2}=\frac{4}{2}=2$	
15	a	Area of the rectangle $=6\times4=24\,cm^2$	1
	b	Area of the sector = $\frac{1}{4}\pi4^2=4\,\pi cm^2$	1
	c	Area of the shaded portion = $(24-4\,\pi)\,cm^2$	2
16	a	Base perimeter = 30 cm	1
	b	LSA = $30\times20=600\,cm^2$	1
	c	$Base\,area=\frac{1}{2}\times5\times12=30\,cm^2$ $TSA=600+30+30=660\,cm^2$	2
17	a	Let x and y be the length and breadth of the rectangle $(x-1)(y-1)=99$ $(x+1)(y+1)=143$ $xy-x-y=98$ $xy+x+y=142$ $2xy=240$ $xy=120$	2
	b	$x+y=22$ Solving x= 12 , y = 10	2
18	a	120^0	
	b	Perimeter of hexagon = 36 cm	
	c	Perimeter of the shaded region $=36+\frac{6\times120}{360}\times2\,\pi\times3$ $=36+12\,\pi$	
19	a	Missing numbers are 10 and 25	2
	b	$x\times y=100$	1

	c	The numbers are in inverse proportion	1
20			1
21	a	Number of members = 40	1
	b	Mean age = $\frac{1920}{40}=48$	3
22	a	$\angle$BAC = 60^0	1
	b	Area of the unshaded sector = $\frac{60}{360}\times\pi\times6^2=6\pi cm^2$	1
	c	Area of the shaded region =Area of the major arc + Area of the triangle – Area of the sector = $\frac{300}{360}\times36\pi+\frac{\sqrt{3}\times10^2}{4}-6\pi=24\pi+25\sqrt{3}$	3
23	a	Diameter = 12 cm , radius = 6 cm	1
	b	Volume of the cylinder	1

		$= \pi \times 6^2 \times 25 = 900\,\pi cm^3$	
	c	Volume of the square prisim $= 12^2 \times 25 = 3600\,cm^3$ Volume of the remaining wax $= 3600 - 900\,\pi = 774\,cm^3$ Volume of the cube = $(1)^3 = 1$ Number of cubes = $\frac{774}{1} = 774$	1 1 1
24	a	P(1) = P(2) = 0 P(0) = 4	1 1
	b	$P(x) = ax^2 + bx + c$ c= 4 a+b = - 4 2a+ b = -2 Solving a = 2 ; b= -6 $P(x) = 2x^2 - 6x + 4$	1 1 1
25	a	LSA $= 4 \times a \times h = 4 \times .04 \times 4 = 6.4\,m^2$	1
	b	Total Surface area of 15 pillars $= 15 \times 6.4 = 96\,m^2$ Cost of painting = $96 \times 90 = Rs\,8640$	2 2

26			5
27	a	$\angle BCQ = 45^0$	1
	b	AC = 8 cm	1
	c	CB = 4 cm	1
	d	Area of the shaded region $= \frac{1}{4}\pi \times 8^2 + \frac{1}{8}\pi \times 4^2$ $= 16\pi + 2\pi$ $= 18\pi$	2
28	a	Median consumption = $\frac{4600}{20}$ = 230 units	5
29	a	$(\frac{1}{4}+\frac{1}{5}) \div (\frac{1}{4}-\frac{1}{5}) = 9$	1
	b	$(\frac{1}{15}+\frac{1}{16}) \div (\frac{1}{15}-\frac{1}{16}) = 31$	1
	c	$(\frac{1}{25}+\frac{1}{26}) \div (\frac{1}{25}-\frac{1}{26}) = 51$	1
	d	$P = \frac{1}{30}$	1
	e	$(\frac{1}{n}+\frac{1}{n+1}) \div (\frac{1}{n}-\frac{1}{n+1}) = 2n + 1$	1